Measures of position

I) Quartile →

[10, 20, 40, 60, 80, 70, 30, 90, 100, 50]

Q_1 = 25-/ Q_3 = 75-/

Q_2 = 50

Q_2 (median) = $\frac{50+60}{2}$ = 55

[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

↑ Q_2

Q_1 = 10, 20, 30, 40, 50

↑ 25-/

Q_3 = 60, 70, 80, 90, 100

↑ 75-/

II) IQR (Inter quartile Range)

$$IQR = Q_3 - Q_1$$
$$= 75 - 25 = 50$$

III) Percentile →

[5, 6, 8, 8, 9, 11, 12, 12, 13, 14, 16, 18, 19, 22]

$$\Rightarrow \frac{9}{14+1} \times 100 \Rightarrow 60\%$$

$$\Rightarrow \frac{11}{15} \times 100 \Rightarrow 73.33\%$$

Formula => $\frac{L + 0.5 \times E}{N} \times 100$

Less than

Equal

N

12th $\quad \frac{6+0.5\times2}{14}\times100$

$\Rightarrow 50$

IV) Percentile Rank →

[4, 5, 6, 8, 9, 11, 12, 13, 15, 16, 18, 19, 22, 23, 24]

80% $\Rightarrow \frac{80}{100}\times(n+1) = \frac{80}{100}\times15+1$

$= \frac{8}{10}\times16 = 12.8$

13th index

22

V) Z-score –

$$\frac{x-\mu}{\sigma}$$

[10, 20, 30, 40, 50]

Zscore of 40 ✓

$x = 40$

$\mu = 30$

$$\Rightarrow \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}}$$

10	30	-20	400
20		-10	100
30		0	0
40		10	100
50		20	400

$$\frac{1000}{5} \Rightarrow \sqrt{200}$$

$\sigma = 14.14$

$$\Rightarrow \frac{x-\mu}{\sigma} = \frac{40-30}{14.14}$$

$Z \Rightarrow \underline{0.70} \Rightarrow \underline{\underline{70\%}}$

$z =$

$z >$

$z <$

$$\begin{bmatrix} \mu = 0 \\ \sigma = 1 \end{bmatrix}$$

Five number summary

max

min

median

Q_1

Q_3

[2, 4, 5, 6, 7, 9, 10, 11, 13, 15, 18, 90]

$$\text{median} = \frac{19}{2} = 9.5$$

$$Q_1 = \frac{25}{100} \times (12+1) = 3.25 = 5$$

$$Q_3 = \frac{75}{100} \times (12+1) = 9.75 = 15$$

$$IQR = Q_3 - Q_1 = 15 - 5 = 10$$

$$\max(\text{upper}) = Q_3 + 1.5\,IQR$$

$$= 15 + (1.5 \times 10) = 30$$

$$\min(\text{lower}) = Q_1 - 1.5\,IQR$$

$$= 5 - (1.5 \times 10) = -10$$

